OPIM639 · Innovations & Operations of Intelligent Online Marketplaces

Chagee: opening overseas stores right

Predict demand, provision the store, size the counter

Group 3

Keziah, Jin, Ayumi, Revan



CONTENTS

Six chapters: from Chagee's problem to a playbook of triggers

1	**Context** Chagee, the processes AI can improve, and the problem	slide 1	0:50
2	**Scope** Which course topics we use, which we leave out, and why	slide 2	0:40
3	**Predict** Read a new market's demand before the store opens	slide 3	0:40
4	**Provision** How much fresh milk to open with	slide 4	0:30
5	**Perform** How a counter absorbs a surge, and which counter to open with	slides 5 and 6 and 7	2:20
6	**Decide** The playbook of triggers, and the size of the benefit	slides 8 and 9	1:50

About 7:00 on stage, then 3:00 for Q&A. Click a chapter to jump to it. Appendix slides are in the handout only.

Chagee plans about 200 overseas openings a year, and each one commits its stock and its counter before its demand is known

A premium tea chain scaling abroad

7,453 stores at end-2025, 345 of them overseas in 7 markets. FY2025 GMV RMB 31.58bn, net revenue RMB 12.91bn. Orders run through Chagee's own app (our understanding).

Three core processes decide a new store

Forecasting its demand, provisioning its opening stock, and setting up its counter. Sourcing is fixed: Chagee says it sources tea and milk directly, with distribution run from HQ.

The problem

Stock and counter capacity are committed before a new market's demand is known, so a miss shows up as wasted milk, missed cups, queues or machines that sit idle (our reading).

Why it matters

The same decision repeats about 200 times a year. Opening crowds can mislead: the Kuala Lumpur store reportedly sold 1,000+ cups on its first day (secondary source).

INTERACTIVE EXPLAINER WHERE AI ENTERS A NEW STORE'S OPENING

Click a step. Today's practice is our reading of public sources and one industry conversation, not a Chagee statement.

1 Members → 2 Forecast → 3 Opening stock → 4 Counter → 5 After opening

STEP	TODAY (OUR READING)	WHAT AI ADDS	WHO DECIDES · WHERE
1 · Members	Members order through the app (our understanding).	**Read sign-ups and pre-orders as early demand evidence.**	Membership / CRM *slide 3*
2 · Forecast	Judgement-led by a local General Manager (our reading).	**A proposed demand range from member signals, with hype and habit separated by existing-market proxies.**	Market-entry leads and GMs *slide 3*
3 · Opening stock	Central, standardised supply of tea and milk.	**Size the first fresh-milk order to that range with newsvendor logic.**	Supply chain / planning *slide 4*
4 · Counter	Moving toward semi-automation: QR-scan dispensing, Chagee Express.	**Simulate the counter before the store exists, and set triggers for adding machines.**	Store format / equipment; franchisees *slides 5 to 7*
5 · After opening	Not covered by public sources.	**Compare actual demand with the range to narrow the next market's forecast.**	Market-entry leads *slide 8*

Printed with all five steps shown. Live: click a step to see one at a time.

THE PROBLEM WE SOLVE

Every overseas opening commits stock and counter capacity before its demand is known.

We do not train an AI model here. We show where AI fits, and test the opening decisions with a simulation built on labelled assumptions.

A closed app-to-store channel puts our problem in Topics 3 to 5, not the marketplace topics

INTERACTIVE EXPLAINER EIGHT COURSE TOPICS: CLICK ONE TO SEE WHY IT IS IN OR OUT

TOPIC 1
~~Introduction to Online Retailing~~
OUT
One closed channel, app to store: no online-versus-store trade-off to weigh (our reading).

TOPIC 2
~~Online Marketplaces~~
OUT
No third-party sellers, so no sell-versus-fulfilment-service choice (our reading).

TOPIC 3
Analytics and AI for Online Retailing
IN
→ Predict · slide 3
Predicting a new market's demand from app-member signals.

TOPIC 4
Supply Chain Management using AI and Robotics
IN
→ Provision · Perform
Supply-chain planning sizes the stock; equipment selection sizes the counter.

TOPIC 5
Order Fulfillment for Online Retailing
IN
→ Perform · Provision
Picking with humans and robots: the bucket brigade behind our counter model.

TOPIC 6
Last-mile Delivery for Online Retailing
NOT USED
Outside our question, which stops at the counter.

TOPIC 7
Service-oriented platforms
NOT USED
Outside our question.

TOPIC 8
~~Omni-channel Retailing~~
OUT
One channel, so no channel conflict; the franchise revenue share is out of scope.

INTERACTIVE EXPLAINER WHERE DOES UNCERTAINTY LAND?

Why this slice: with sourcing fixed, a demand miss lands in the store, on its opening stock or its counter speed (our reading, not a Chagee statement).

Rigid sourcing | Localised sourcing

Rigid sourcing Chagee today, our reading

Sourcing	○○○	Fixed: single origin
Distribution	○○○	Fixed: run from HQ
Opening stock	●●●	Absorbs the miss
Counter speed	●●●	Absorbs the miss

Localised sourcing For contrast only

Sourcing	●●○	Flexes with local suppliers
Distribution	●○○	Shorter, local
Opening stock	●○○	Less exposed
Counter speed	●●●	Still exposed

●○○ low ●●● high exposure to a demand miss

Printed: every tile shows its status and reason; both sourcing columns are shown.

OUR SCOPE

Three topics, three decisions.

Topic 3 feeds Predict; Topics 4 and 5 feed Provision and Perform. Sourcing is fixed by design, so we take it as given, and the franchise revenue share stays out of scope.

App-only ordering could let Chagee read a new market's demand before opening; we propose the method and test how much it matters

Every order is tied to a member

With app-only ordering (our understanding), each cup is linked to a member account, so a pre-launch waitlist could be read as a demand signal.

A narrower band is our assumption

The slider narrows the band from ±50% to ±15% along a straight line we assumed. We have not shown that member data does this: read it as a sensitivity.

Hype versus habit needs a proxy

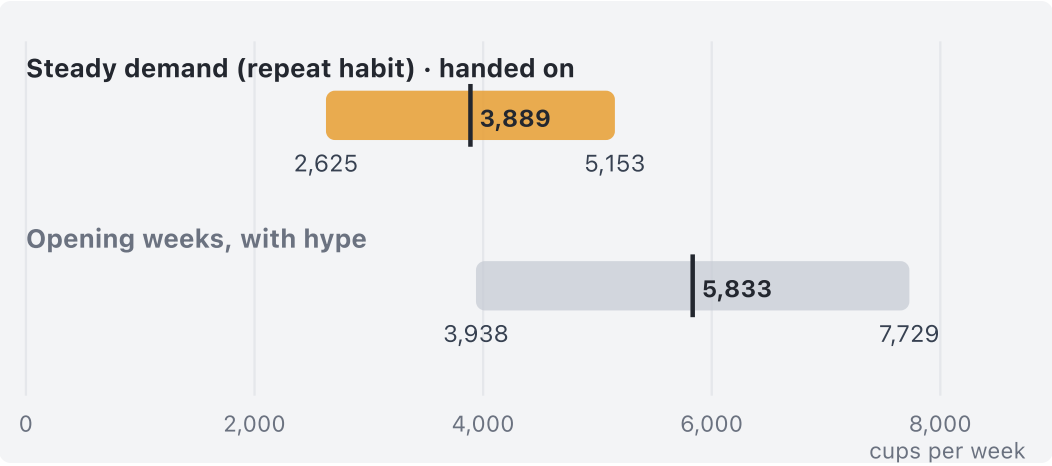
A new store has no repeat history. Proxies: repeat rates in Chagee's existing overseas markets, nearby stores, and waitlist members who already order elsewhere. Kuala Lumpur reportedly sold 1,000+ cups on day one (secondary).

Blind spot: tourists and non-app orders

Visitors and tourist exceptions sit outside the member signal, so the forecast understates demand where tourists matter. We name this gap rather than model it.

INTERACTIVE EXPLAINER DEMAND RANGE BEFORE OPENING: A SENSITIVITY

Weekly cups at one new store. The band covers 80% of likely outcomes; how fast it narrows with signal is our assumption.



Assumed signal strength — 50% → ±32.5%

Opening hype removed | Hype included

Printed at the default assumed signal strength (50%) with opening hype removed.

BAND HANDED ON

±32.5% of base

Low	2,625
Base	3,889
High	5,153
Width	2,528

Cups per week. Opening hype removed: steady demand.

Not in the signal: tourists and non-app orders, so true demand is likely higher.

PROPOSED METHOD

A method to test, not a result.

Back-test it: forecast past openings from their pre-launch sign-ups, and check the band holds about 80% of actual demand. To our knowledge, reading app-only member data this way before launch is new.

DECISION STRIP

DECISION
Does the forecast band change which counter a new store opens with?

WHO DECIDES
Market-entry leads and GMs; Membership / CRM runs the signal

TRIGGER · COMPUTED BY SIM/
±57% band half-width. Below it all-human is cheaper, above it mixed (swept ±0% to ±60%).

Mixed vs full automation: No break-even in range (swept ±0% to ±60%): mixed is cheaper throughout, so this choice holds everywhere.

ACTION
Use the band for stock, not the counter: on the band alone mixed wins only past the trigger, and full automation never does. The counter turns on volume and input costs.

Now ±32.5%: below the trigger, so on the band alone all-human is cheaper.

The wider the demand range, the costlier the opening-stock decision, especially with fresh milk

Two ways to be wrong

Too much fresh milk expires; too little turns customers away. Newsvendor logic: order until one more litre's expected waste cost equals the expected lost-sale cost it avoids.

Here a stockout costs more than waste

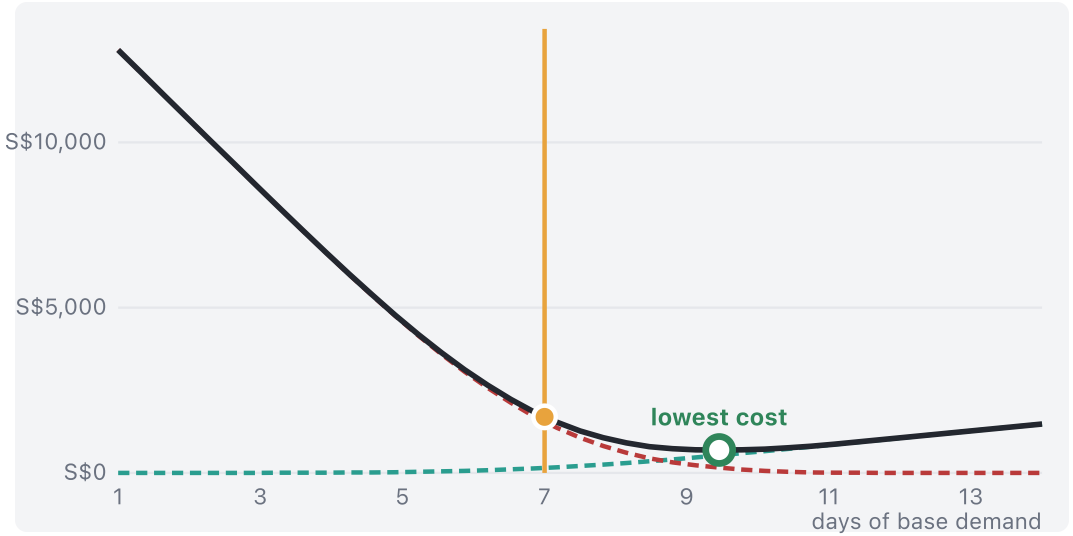
A missed cup loses S$3.82; a spare litre of milk wastes S$2.55. So the best order sits high in the band, at the 91% point of demand.

Shelf life caps the buffer

Fresh milk keeps about 7 days (illustrative), so extra stock cannot be carried forward. A narrower band from slide 3 is the cheaper fix.

INTERACTIVE EXPLAINER OPENING STOCK CALCULATOR

Expected cost of the first fresh-milk order at one store, using slide 3's band. Overseas milk sourcing is unknown; figures are generic.



Total expected cost · Milk wasted · Cups missed

Days of opening stock 7 days

Printed at the band from slide 3 and the default of 7 days of stock.

AT 7 DAYS OF STOCK

S$1,695 expected

Milk wasted	**S$149**
Cups missed	**S$1,546**

BEST ORDER AT THIS BAND (±32.5%)

9.5 days	**S$694**
Same order, strong signal	**S$320**

Hand-off: open with 788 L of milk. The counter is decided by volume and input costs, not this band (slide 7).

OUR VIEW

Narrow the band before you stock the store.

From no signal to the default signal, the expected cost of the best opening order falls from S$1,064 to S$694 per store (illustrative). The band matters for stock; for the counter, volume and input costs decide (slide 7).

DECISION STRIP

DECISION

How much fresh milk does a new store open with: above or at the base forecast?

WHO DECIDES

Supply chain / planning, from the market-entry team's demand band

TRIGGER · COMPUTED BY SIM/

S$25.47/L waste cost per litre at which the best order falls to the base forecast. = S$3.82 per lost cup ÷ 0.15 L per cup. Above the whole S$1.27 to S$5.09/L range, so ordering above base holds everywhere.

ACTION

Order above the base forecast, at the cost-minimising level the calculator marks. Narrow the band first: it lowers the cost of even the best order.

Now ±32.5%: best order 9.5 days, about 788 L of milk.

In a surge, people pass work along like a bucket brigade; machines cannot, so the slowest station sets the pace

People pass the work along

Each worker carries a drink downstream and hands it over. In a surge the handover point moves, so the load evens out: the Week 5 bucket-brigade result.

Machines stay at their station

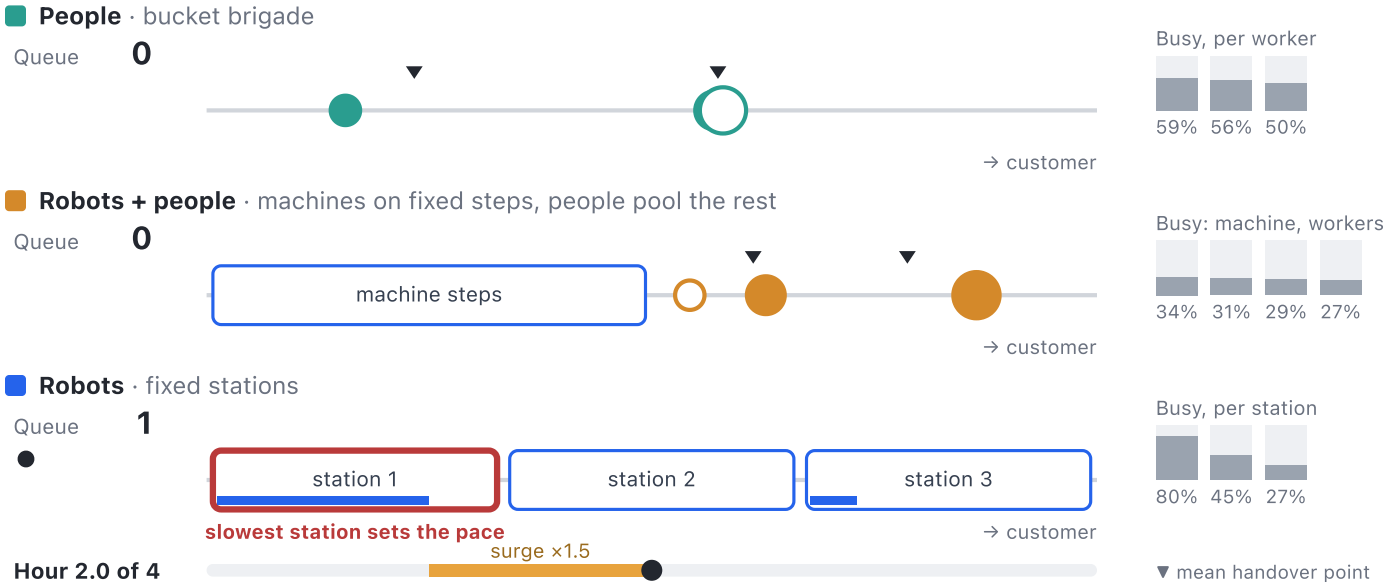
A fixed station cannot hand work back or take on more. The queue builds at the slowest one, wherever it sits on the line.

Sized for the peak, machines sit idle

Switch to as sized in our cost model: the robots stop queuing, but they are busy only part of the time. Idle capacity is what a store pays for.

INTERACTIVE EXPLAINER INSIDE THE STORE: A BUCKET BRIGADE AT THE COUNTER

Mechanism illustration on the Week 5 bucket-brigade engine, in its units. Times and costs come from the next two slides, not from this animation.



Moderate ×1.5 surge, 4-hour run	Machines matched to crew		As sized in cost model	
	Peak queue	Busy (range)	Peak queue	Busy (range)
People	0	57%–66%	0	57%–66%
Robots + people	0	28%–32%	0	10%–33%
Robots	2	38%–90%	1	19%–40%

Printed at the end of the surge window; the table compares both sizing settings over the full run.

OUR VIEW

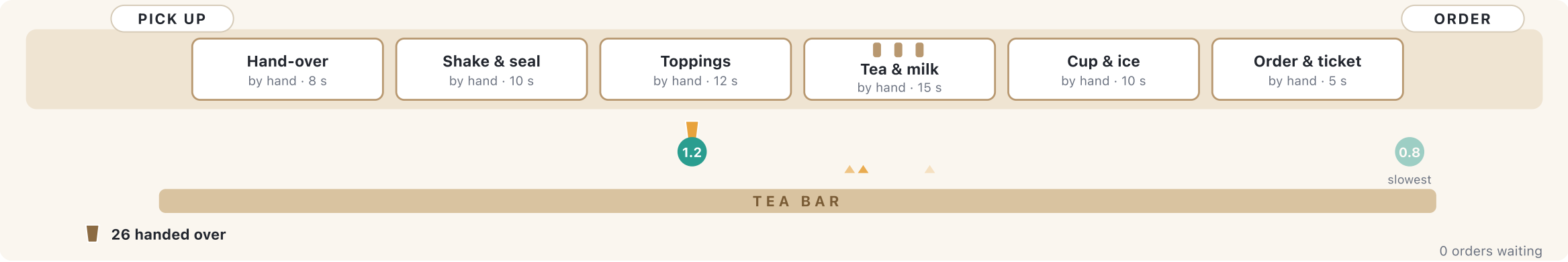
People flex; machines come in whole units.

So the counter choice turns on volume. The next two slides measure it: time to serve, then cost.

In the peak hour, robots hand over a cup in about 49 seconds; a counter like today's takes about 2.6 minutes

INTERACTIVE EXPLAINER STORE REPLICA: ONE COUNTER IN THE PEAK HOUR

Illustrative layout after a Chagee tea bar: order right, brewing behind the counter, pick-up left. People move as the Week 5 bucket brigade; times, staff and arrivals from the cost model.



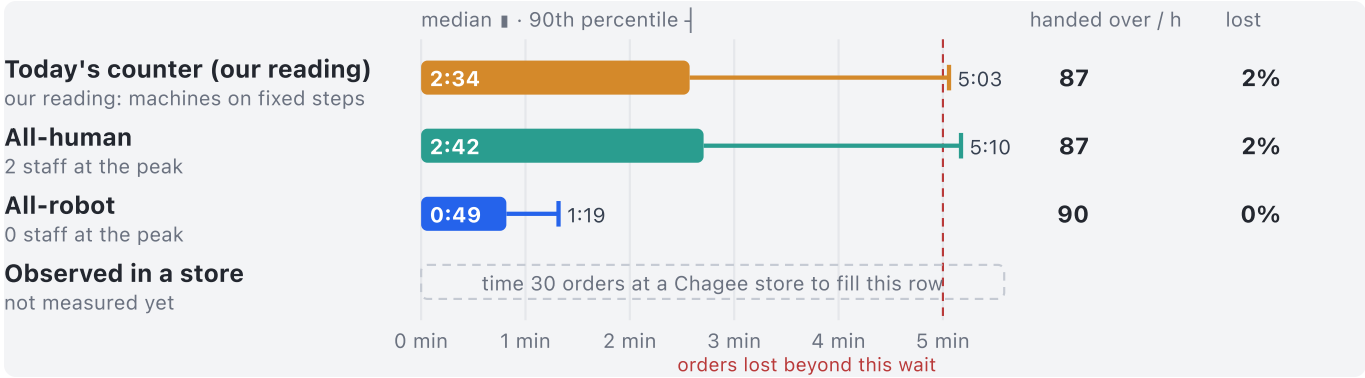
Minute 20 of the peak hour Handovers **30** 0.8× working **79%** 1.2× working **88%** *Animation of the movement; times and lost orders below come from the cost model.*

Number = worker speed (slowest at the order point) · cup = drink in hand · faded = walking back · red ring = blocked · ▲ = handover. Robots stay put; cups move.

Printed at minute 20 of the peak hour: all-human counter, moderate surge.

INTERACTIVE EXPLAINER TIME TO SERVE: TODAY VERSUS SIMULATED

From order to hand-over, waiting included. Cost model, peak hour, 20 simulated days per bar.



Printed at the moderate surge.

How the people move

Week 5 bucket brigade: carry a cup to pick-up, walk back, take over from the slower colleague behind. Robots stay put; cups move.

Where the numbers come from

Times and lost orders come from the cost model, so they match the next slide. Today's counter is our reading, not measured.

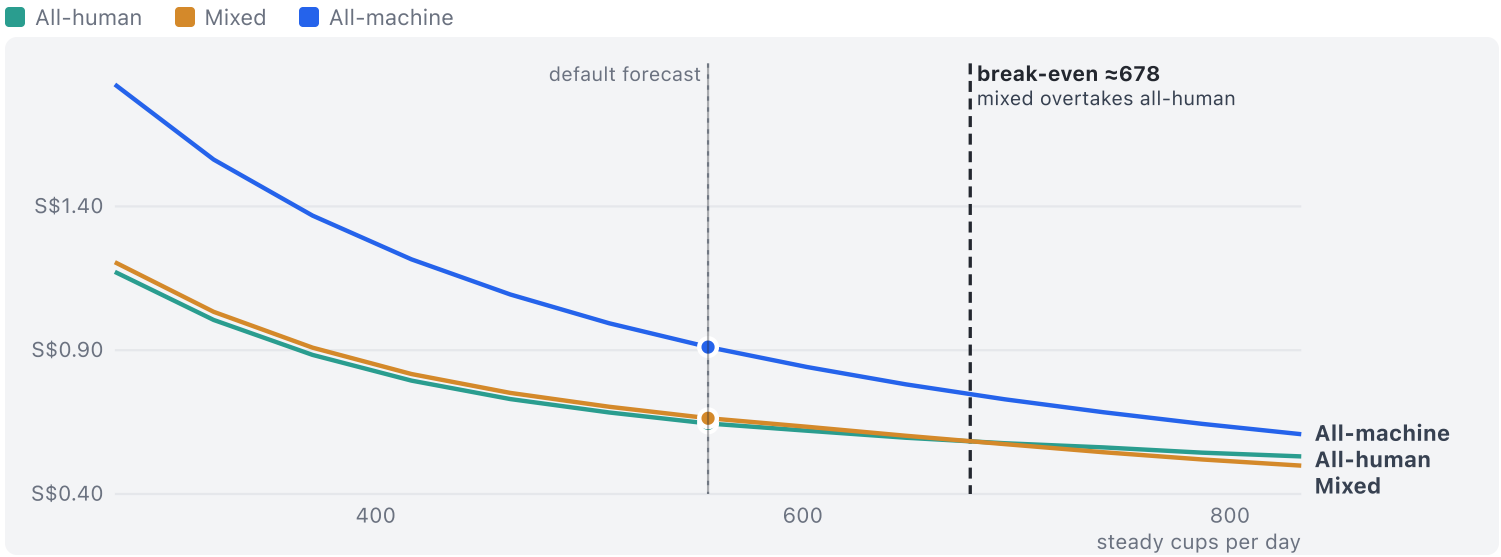
OUR VIEW

Robots buy speed in the peak; whether it pays is the next slide.

All-human and mixed are S$0.02 a cup apart at our forecast; mixed wins once volume, wage or machine cost crosses a trigger

INTERACTIVE EXPLAINER COST PER CUP, BY COUNTER

Cost model: labour, machines and lost sales per cup served; default band, moderate surge.



All-human	Mixed	All-machine
S$0.65	**S$0.66**	**S$0.91**
cheapest	+S$0.02	+S$0.27

At 556 cups/day. All-robot machines are busy 21% of open hours.

Printed at the default forecast.

WHEN DOES MIXED BEAT ALL-HUMAN?

MIXED WINS WHEN…	IS	OURS	GAP
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share	none in 0% new staff to 50% new staff: all-human cheaper throughout		
Machine time	none in 0.70× to 2.00×: all-human cheaper throughout		

Break-evens from sim/; nearest trigger first.

How sure are we? A close call. The gap is S$0.02 a cup. Our staffing shortcut misses its Week 5 check (4.7% and 6.8% against ±3%), and with walking modelled people reach only 90% of the assumed capacity (70% at today's counter). Test the ranking with real staffing data before relying on it.

DECISION STRIP

DECISION
Which counter does a new store open with, and when are machines added?

WHO DECIDES
Store format / equipment, with franchisees (capex and labour)

TRIGGER · COMPUTED BY SIM/
678 cups/day steady cups per day. Below it all-human is cheaper, above it mixed (swept 278 cups/day to 3,000 cups/day). Trigger range, not a new-store forecast.

Full automation: 1,820 cups/day · step.

ACTION
No default: all-human while a store sits below every trigger, mixed once any is crossed, full automation only past about 1,820 cups a day.

Now 556 cups/day: below the volume trigger, so on volume alone all-human is cheaper.

The playbook: no default counter; a few measurable triggers decide each opening

TRIGGER	FLIPS WHEN	OURS	GAP
Counter: when mixed beats all-human · Store format / equipment, with franchisees			
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share	none in 0% new staff to 50% new staff: all-human cheaper throughout		
Machine time	none in 0.70× to 2.00×: all-human cheaper throughout		
Counter: when full automation beats mixed · Store format / equipment, with franchisees			
Steady volume	above **1,820 cups/day** · step	556 cups/day	+228%
Wage per hour	none in S$14.09/h to S$25.45/h: mixed cheaper throughout		
Machine cost	none in S$636/month to S$3,818/month: mixed cheaper throughout		
Forecast band	none in ±0% to ±60%: mixed cheaper throughout		
Machine time	none in 0.70× to 2.00×: mixed cheaper throughout		
Stock: when to order at or below the base forecast · Supply chain / planning			
Milk waste per litre	none in S$1.27 to S$5.09/L: order above base throughout (flips at S$25.47/L = S$3.82 per lost cup ÷ 0.15 L)		

Break-evens from sim/, each with every other input at our defaults; nearest first. The all-human vs mixed gap is S$0.02 a cup and rests on a simplified staffing model (slide 7).

INTERACTIVE EXPLAINER YOUR MARKET

Set your inputs. Verdict and margin: the full cost model (500 simulated stores; under S$0.01 a cup is a tie). Nearest trigger: a live search with 20.

Wage per hour		S$15.06
Machine cost, per unit per month		S$1,909
Expected steady cups per day		556
Forecast band (±)		±32.5%
Milk waste cost per litre		S$2.55

COUNTER **All-human** cheaper than Mixed by S$0.02 a cup

All-human **S$0.65** Mixed **S$0.66** All-machine **S$0.91**

Nearest trigger (live estimate): Mixed wins if machine cost, per unit per month falls to about **S$1,783** (7% away).

Stock: order above the base forecast (best order at the 91% point of demand); flips only if waste costs over S$25.47 a litre.

Printed at our default inputs.

OUR VIEW

Measure the triggers, then choose the counter.

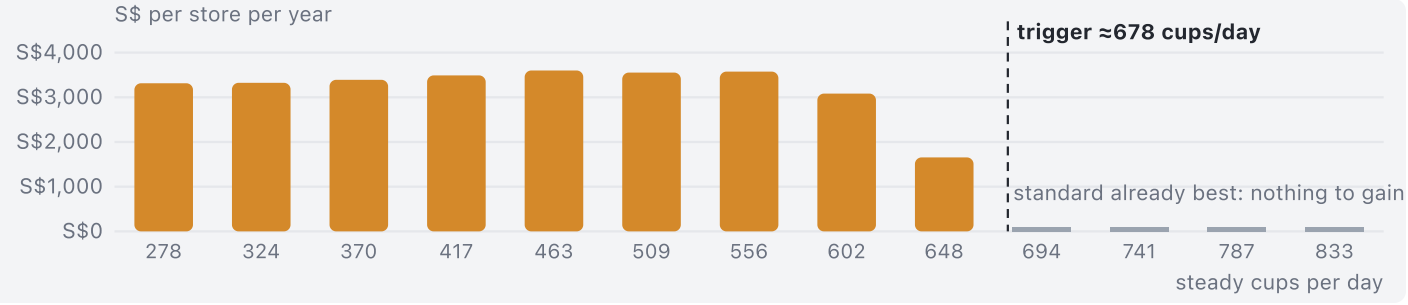
Before each opening: price the machine, set the wage, and back-test the volume forecast. The nearest trigger tells you what to check first.

Choosing each counter by trigger, instead of one standard, is worth about S$0.5m a year: small per store, and nothing where the standard already wins

INTERACTIVE EXPLAINER WHAT CHOOSING BY TRIGGER IS WORTH

Compared with: Chagee's semi-automated counter (our mixed setup) at every new store. This is a modelled avoided cost, not a demonstrated saving.

Per store per year: the standard's cost above the cheapest counter at that volume (default band, moderate surge).



Overseas openings a year 200

Share of openings below the trigger 72%

Default share: openings spread evenly over our volume range (illustrative).

≈ S$0.5m

a year, illustrative

200 openings × (**72%** below the trigger × **S$3,220** per store + **S$370** stock cost avoided)

≈ S$0.5m from choosing the counter + S$0.1m from opening stock, if the pre-launch signal works as assumed (slide 3). For reference: had the standard been full automation, it would cost S$32,724–S$65,190 per store a year more than the cheapest counter.

Printed at our defaults: openings spread evenly over our volume range, so 72% sit below the trigger.

RISKS

1. **Staff dependence and turnover**
   A people-based counter relies on trained staff, and new hires are slower. In our range turnover did not flip the choice (slide 8), but real staffing data could (slide 7).
2. **Tourist and non-app demand is invisible**
   The member signal misses visitors, so the band sits too low in tourist areas. Widen it there and log walk-up exceptions after opening.
3. **Opening crowds read as steady demand**
   Sizing to opening-week queues overbuys machines and staff. Separate first orders from repeat orders before committing capex.
4. **Automation versus the premium teahouse**
   Heavy automation could undercut the premium experience. The mixed setup keeps people at the customer-facing steps: finishing and handover.

Next step with real Chagee data

- Replace every assumption with store actuals: station times, staffing, wages, machine costs.
- Back-test the waitlist signal on recent openings, such as Korea (launch planned for Q2 2026).
- Pilot all-human against mixed counters in paired new stores, timing real orders.

OUR CONCLUSION

The value is in choosing per store, not in one standard.

Uncertainty did not decide the counter, so our starting claim fell. Read demand before opening for stock; choose the counter on measured volume, wage and machine cost. To our knowledge, simulating a store before it exists is new.

Every assumption, with its unit, range and source

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Facts				
stores end 2025	7,453	stores	–	L-jiemian
overseas stores	345	stores	–	L-jiemian
overseas markets	7	markets	–	L-jiemian
gmv 2025 rmb bn	31.58	RMB bn	–	L-jiemian
net revenue 2025 rmb bn	12.91	RMB bn	–	L-jiemian
tea latte share china	0.91	share of China sales	–	L-prometheus
kl opening day cups	1,000	cups (reported, '1,000+')	–	L-gabgrowth
Predict				
signal strength default	0.5	share, 0 = no pre-launch signal, 1 = strong signal	0 to 1	assumption
Predict Toggles				
remove hype default	yes	boolean	–	assumption
Store				
open hours per day	12	hours/day	–	assumption
open days per year	360	days/year	–	assumption
cups per store per year	200,000	cups/store/year	100,000 to 300,000	assumption
hourly demand share	0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07	share of daily cups per open hour (10:00 to 22:00), sums to 1	–	assumption
Demand				
uncertainty definition	half_width_pct	text	–	assumption
band coverage	0.8	probability	–	assumption
demand floor share of base	0	share of base	–	assumption
uncertainty no signal	0.5	half-width, share of base	0.35 to 0.7	assumption
uncertainty strong signal	0.15	half-width, share of base	0.05 to 0.25	assumption
opening hype multiplier	1.5	x steady-state demand	1 to 2.5	assumption
opening hype weeks	2	weeks	1 to 6	assumption
uncertainty sweep	from 0 · to 0.6 · step 0.05	half-width, share of base	–	assumption
Surge				
surge multiplier	none 1 · moderate 1.5 · large 2.5	x forecast arrival rate	–	assumption
surge hours	2	busiest forecast hours that get the surge	–	assumption
surge default	moderate	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Stations				
list	ticket, cup_ice, tea_milk, topping_finish, shake_seal, handover	text	–	assumption
human seconds per cup	ticket 5 · cup ice 10 · tea milk 15 · topping finish 12 · shake seal 10 · handover 8	seconds of one average worker per cup	0.8 to 1.25	assumption
machine seconds per cup	ticket 2 · cup ice 8 · tea milk 10 · topping finish 15 · shake seal 8 · handover 6	seconds per cup per machine unit	0.7 to 2	assumption
mixed machine stations	ticket, tea_milk, shake_seal	text	–	assumption
machine speed variation cv	0	coefficient of variation	–	assumption
People				
human speed variation cv	0.25	coefficient of variation, minute to minute	0.1 to 0.4	assumption
worker speed spread	0.2	+/- share around average speed	0 to 0.4	assumption
coordination loss human	0.9	multiplier on pooled throughput	0.75 to 1	assumption
coordination loss mixed	0.9	multiplier on human-pool throughput in the mixed setup	0.75 to 1	assumption
new staff share	0.25	share of roster in first months	0 to 0.5	assumption
new staff relative speed	0.7	x average speed	0.5 to 0.9	assumption
min staff on shift	all human 2 · mixed 1 · all machine 0	workers per open hour	–	assumption
all machine attendant per open hour	1	workers per open hour	0.5 to 2	assumption
staffing flexibility	0.7	share of roster gap closed after demand is known	0.3 to 1	assumption
Costs				
labour cost per hour	15.06	SGD per hour worked, fully loaded (statutory costs)	14.09 to 25.45	L-mom-pwm, L-mom-ows, L-cpf, L-sdl, L-mom-leave, L-mom-ph
machine cost per unit per month	1,909	SGD per station machine unit per month	636.2 to 3,818	assumption
lost order cost	3.82	SGD per lost order	1.91 to 7.64	assumption
capacity safety margin	1.2	x forecast peak rate	1 to 1.5	assumption
Service				
target wait minutes	5	minutes	2 to 10	assumption
Provision				
milk litres per cup	0.15	litres per cup	0.1 to 0.25	assumption
milk waste cost per litre	2.55	SGD per litre thrown away	1.27 to 5.09	assumption
milk shelf life days	7	days	3 to 14	assumption
stockout cost per cup	3.82	SGD per cup not served for lack of stock	1.91 to 7.64	assumption
opening stock days	7	days of forecast base demand	1 to 14	assumption
Benefit				
overseas stores per year	200	stores/year	50 to 300	L-jiemian
saving per cup	from_simulator	SGD/cup	–	assumption
waste avoided	from_provision	SGD/store/year	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Simulation				
monte carlo draws	500	demand draws per uncertainty point	–	assumption
base seed	639	integer	–	assumption
minute step	1	minutes	–	assumption
live draws	20	simulated stores per live estimate	–	assumption
Triggers				
volume trigger range cups per day	277.8, 3,000	cups per store per day	–	assumption
Illustration				
worker speeds	1, 1.5, 2	lines per hour	–	assumption
walk back speed	6	lines per hour	–	assumption
base load share	0.6	arrival rate as share of the crew's nominal capacity (sum of speeds)	–	assumption
machine sizing	matched station rates = 3 x worker speeds (each station does a third of the work; same total speed as the cr…	text	–	assumption
mixed sizing	matched machine block rate = the people part's nominal rate (crew speed / hand share), a balanced mixed line…	text	–	assumption
machine buffer capacity	2	orders between consecutive machine stations	–	assumption
run hours	4	simulated hours per run	–	assumption
surge window hours	1, 2	start and end hour of the surge within a run	–	assumption
Validation				
breakeven	tol cost sgd per cup 0.01 · tol x share of range 0.005 · grid points 13	mixed	–	assumption
week5 comparison	tolerance pct 3 · engine Brigade() read from the ENGINE string in week5/build_opim639_deck.py, not modified …	mixed	–	assumption
cost per cup aggregation	ratio_of_sums	text	–	assumption
peak utilisation definition	forecast peak-hour arrival rate x surge multiplier, divided by the setup's nominal capacity in that hour, at…	text	–	assumption
crossover definition	smallest u at which mixed cost/cup <= all-machine cost/cup and stays <= for every larger u in the sweep, lin…	text	–	assumption
crossover stability	seeds 20 · min seeds with crossover 18 · max spread pct points 10	mixed	–	assumption
sensitivity	one-at-a-time: each parameter with a non-null range set to its low then high value, all others at default; r…	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

The model is stable across seeds but misses its Week 5 check, and a handful of inputs decide all-human versus mixed

Week 5 check: pooled staffing formula vs the Week 5 engine

WORKER SPEEDS	ENGINE, LINES/H	POOLED	GAP
1, 1.5, 2	4.300	4.500	**4.65%**
1, 2, 3	5.620	6.000	**6.76%**

Fails the pre-registered ±3% tolerance: the pooled formula overstates a real bucket brigade. The coordination factor (0.9) absorbs part of this.

Seed stability

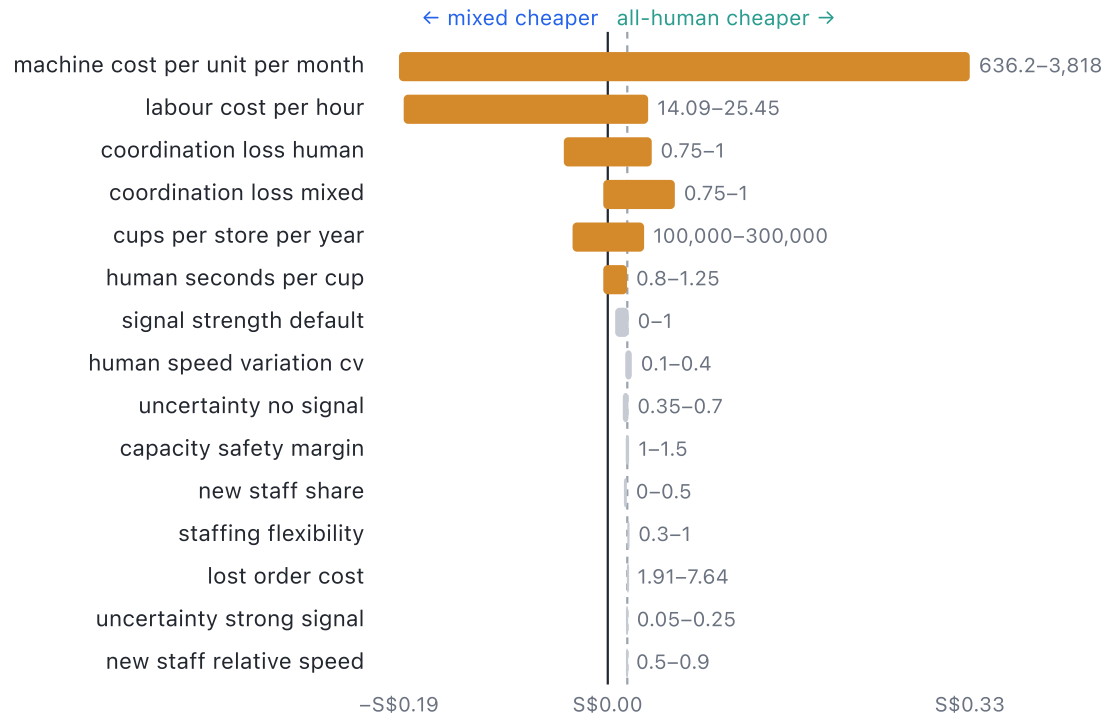
Uncertainty sweep at the moderate surge: 20 of 20 seeds: mixed cheaper than full automation at every band width. The result agrees in every seed. The pre-registered rule asks for a crossover point in most seeds; with none in any seed, the deck shows no crossover point, as the rule requires.

Store replica (walking modelled) vs cost-model capacity, cups per minute

COUNTER	REPLICA	COST MODEL	SHARE
all-human	1.50	1.67	90%
mixed	1.17	1.67	70%
all-machine	3.98	4.00	100%

The replica animates movement only; times and costs on the slides come from the cost model.

Sensitivity: mixed minus all-human cost per cup, one assumption at a time



Dashed: our default gap (S$0.018 a cup, all-human cheaper). Amber bars cross zero inside their approved range, so they can flip the choice; labels show the range tested. No visible effect: target wait minutes, worker speed spread, machine seconds per cup, opening hype multiplier, opening hype weeks, all machine attendant per open hour, milk litres per cup, milk waste cost per litre, milk shelf life days, stockout cost per cup, opening stock days, overseas stores per year.

At a new store's volume, a mixed people-and-machine counter beats full automation, because machines sit idle, not because demand is uncertain

We tested the hypothesis that mixed wins only until demand becomes predictable. Our simulator rejected the uncertainty part.

Machines come in whole units, so they sit idle

Even one machine per station serves about 240 cups an hour against a forecast peak of about 61. The fully automated line is busy only 21% to 22% of open hours.

People flex; machines cannot be returned

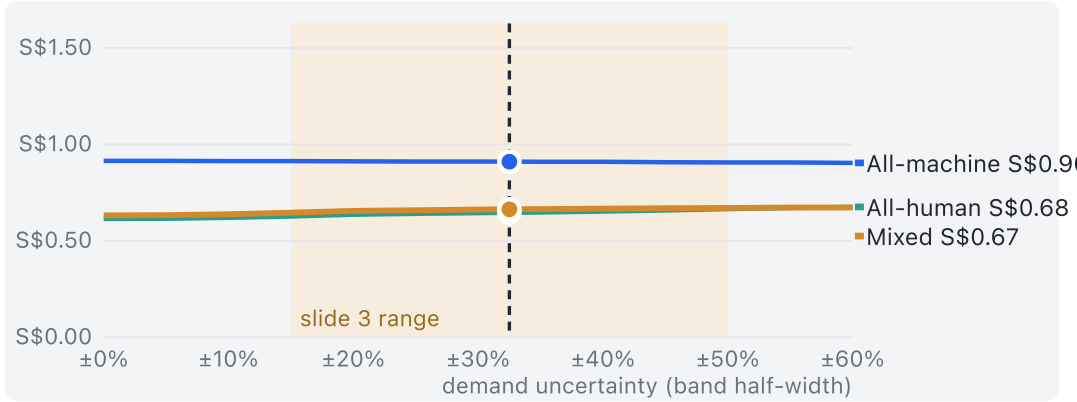
Rosters follow the hour and the real demand, and staff rebalance across stations like a bucket brigade. A mixed counter buys 3 machines instead of 6, with similar staff hours.

Where machines still win

Big surges: at a 2.5× peak, people-based counters lose about 11% of orders and full automation is cheaper. Also steady high volume, consistency and labour scarcity.

INTERACTIVE EXPLAINER COST PER CUP BY COUNTER SETUP

None ×1 surge | **Moderate ×1.5 surge** | Large ×2.5 surge



No crossover: mixed is cheaper than full automation at every uncertainty level shown.

Demand uncertainty ±32.5%

COST PER CUP AT ±32.5%

All-human	S$0.65
Mixed	S$0.66
All-machine	S$0.91
Mixed – machine	–S$0.25

WHAT THE COST IS MADE OF



Automated line busy	21%
Orders lost: mixed	1.3%

What could shift the balance

Machine cost per month	Low S$636	**Default**	High S$3,818	→ mixed cheaper everywhere
Staff turnover (share new)	Low 0%	**Default**	High 50%	→ mixed cheaper everywhere
Forecast narrows (slide 3 band)	No signal ±50%	**Slide 3**	Strong ±15%	

Printed at the default band (±32.5%) and moderate surge.

OUR VIEW

Uncertainty did not decide the counter; volume, wage and machine cost do.

All-human (S$0.65) and mixed (S$0.66 per cup at the default band) are a close call settled by the triggers on slide 7. Machine breakdowns are not modelled, which flatters automation.

The source behind every Chagee fact, and the claims we deliberately did not make

ID	SOURCE AND WHAT IT SUPPORTS	STATUS
L-jiemian	Jiemian (en.jiemian.com), earnings-call coverage: 7,453 stores end-2025; about 200 overseas stores planned for 2026; 345 overseas stores; 7 overseas markets. (brief.md section 9)	PUBLIC
L-gabgrowth	GabGrowth (secondary): first overseas store, Kuala Lumpur, reportedly sold 1,000+ cups on opening day. (brief.md section 9)	SECONDARY
L-unsw	UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)	PUBLIC
L-mom-pwm	Ministry of Manpower, Progressive Wage Model, Food Services sector, https://www.mom.gov.sg/employment-practices/progressive-wage-model/food-services-sector (page updated 1 Sep 2026; fetched 4 Oct 2026). Category A (quick service…	PUBLIC
L-cpf	CPF Board, 'How much CPF contributions to pay', https://www.cpf.gov.sg/employer/employer-obligations/how-much-cpf-contributions-to-pay (page updated 11 Aug 2026; fetched 4 Oct 2026). Employer contribution 17% of wages from 1 Jan…	PUBLIC
L-prn	PR Newswire, 21 May 2024 (company release): Chagee says it sources all tea and milk directly for domestic operations, with distribution managed through HQ. (brief.md section 9)	PUBLIC
L-prometheus	Prometheus Capital Substack (SECONDARY - flag when used): own Yunnan plantations, cold chain to 6,000+ stores; tea lattes about 91% of China sales. (brief.md section 9)	SECONDARY
L-chainreact	Chainreact Substack (secondary): founder avoided fruit tea partly for supply chain complexity. (brief.md section 9)	SECONDARY
L-triunity	Studio Triunity, citing the Mixue prospectus: Chagee's pure milk-tea menu seen as simpler to export; Mixue has 7 SEA warehouses and 3 planned local factories. (brief.md section 9)	PUBLIC
L-cyberrt	Cyber-RT / Citizens Journal, Oct 2022: Chagee Express (Malaysia), semi-automated machine, three drink categories for speed. (brief.md section 9)	PUBLIC
L-gongcha	FastCasual and franchise.org, Jan-Feb 2026 (company-reported): Gong cha Super Wu, up to 65% higher peak-hour productivity, about one minute off prep, stores with as few as two staff, 40-store pilot in 13 countries. (brief.md sec…	PUBLIC
L-starbucks	Constellation Research; Harvard D3: Starbucks Deep Brew - forecasting, scheduling, replenishment, order sequencing; location analytics for new stores. (brief.md section 9)	PUBLIC
L-mosimtec	MOSIMTEC, Winter Simulation Conference 2025: store digital twins replicate existing layouts. (brief.md section 9)	PUBLIC
L-slate	Slate, Aug 2021: bubble tea shaker robot needed about one worker vs four to six; two earlier all-robot shops closed within about a year. (brief.md section 9)	PUBLIC
I-leeyen	INFORMAL - call with Lee Yen. Do not cite without checking with her: Yunnan standard sourcing, GM-led new-market forecasting, app-only ordering with tourist exceptions, sub-one-minute fulfilment target.	INFORMAL
L-xrates	X-Rates monthly averages, https://x-rates.com/average/?amount=1&from=USD&to=SGD&year=2026 (fetched 4 Oct 2026). September 2026 USD to SGD average 1.272467 (30 days). Secondary aggregator; used only for converting the earlier USD…	SECONDARY

Claims we did not make

- That Chagee is first at anything, or that anything is not available in the market.
- Any revenue-mix share for franchise supply: sources conflict.
- Overseas milk sourcing details: unknown, so milk costs are generic.
- A Supply Chain Security Act local-sourcing rule: low-quality source.
- Any Chagee internal cost, forecast method or automation plan not in the ledger.
- Measured serve times: every time on the store-replica slide is modelled; the observed row is empty.
- That member data narrows the forecast band: the narrowing is our assumption, tested as a sensitivity.
- That all-human or mixed is proven cheaper: the gap is small and rests on a simplified staffing model.
- A saving against today's practice: the benefit is a modelled avoided cost against a standard counter.
- Anything about menu prices: that would need a demand-elasticity assumption we do not have.
- The informal sub-one-minute fulfilment target from the industry call: not used.